

MPG No. 37.3.153 $\bar{6}'m'2$ (-6'2m' setting) [Type III, hexagonal] [T tensor]

* Rank 0 tensor. * Rank 1 tensor. * Rank 2 tensor (s). * Rank 2 tensor (a). * Rank 3 tensor (ss). * Rank 3 tensor (sa).

$$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ T_{yzx} & 0 & 0 \\ 0 & -T_{yzx} & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$T_{yzx} = 3M_{du}^{(1)}$$

* Rank 3 tensor (a).

$$\begin{bmatrix} T_{yzx} & 0 & 0 \\ 0 & T_{yzx} & 0 \\ 0 & 0 & T_{xyz} \end{bmatrix}$$

$$T_{yzx} = -M_{du}^{(2)} + M_s^{(1)}$$

$$T_{xyz} = 2M_{du}^{(2)} + M_s^{(1)}$$

* Rank 4 tensor (sss).

$$\begin{bmatrix} 0 & 0 & 0 & T_{xyz} & 0 & 0 \\ 0 & 0 & 0 & -T_{xyz} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ T_{xyz} & -T_{xyz} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & T_{xyz} \\ 0 & 0 & 0 & 0 & T_{xyz} & 0 \end{bmatrix}$$

$$T_{xyz} = T_{gb}^{(1)}$$

* Rank 4 tensor (ssa).

$$\begin{bmatrix} 0 & 0 & 0 & T_{xyz} & 0 & 0 \\ 0 & 0 & 0 & -T_{xyz} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ -T_{xyz} & T_{xyz} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -T_{xyz} \\ 0 & 0 & 0 & 0 & T_{xyz} & 0 \end{bmatrix}$$

$$T_{xyz} = 2M_{f2}^{(1)}$$

* Rank 4 tensor (aas). * Rank 4 tensor (aaa). * Rank 4 tensor (sa).

$$\begin{bmatrix} T_{xyz} & 0 & 0 \\ -T_{xyz} & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & -T_{xyz} & 0 \end{bmatrix}$$

$$T_{xyz} = M_{f2}^{(2)}$$

* Rank 4 tensor (as).

$$\begin{bmatrix} T_{yzxx} & -T_{yzxx} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -T_{yzxx} \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$T_{yzxx} = M_{f2}^{(3)}$$

* Rank 4 tensor (s).

$$\begin{bmatrix} 0 & 0 & 0 & T_{xxyz} & 0 & 0 & T_{xxzy} & 0 & 0 \\ 0 & 0 & 0 & -T_{xxyz} & 0 & 0 & -T_{xxzy} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ T_{yzxx} & -T_{yzxx} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & T_{yzxx} & 0 & 0 & T_{yzxx} \\ 0 & 0 & 0 & 0 & T_{xxzy} & 0 & 0 & T_{xxzy} & 0 \end{bmatrix}$$

$$T_{xxyz} = 2M_{f2}^{(1)} + M_{f2}^{(2)} + T_{gb}^{(1)}$$

$$T_{xxzy} = 2M_{f2}^{(1)} - M_{f2}^{(2)} + T_{gb}^{(1)}$$

$$T_{yzxx} = -2M_{f2}^{(1)} + T_{gb}^{(1)}$$

* Rank 4 tensor (a).

$$\begin{bmatrix} T_{yzxx} & -T_{yzxx} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -T_{yzxx} & 0 & 0 & -T_{yzxx} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$T_{yzxx} = M_{f2}^{(3)}$$

* Rank 4 tensor (t).

$$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & T_{yyyy} \\ 0 & 0 & 0 \\ 0 & T_{yyyy} & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -T_{yyyy} \\ 0 & 0 & 0 \\ 0 & -T_{yyyy} & 0 \\ 0 & 0 & 0 \\ -T_{yyyy} & 0 & 0 \end{bmatrix}$$

$$T_{yyyy} = -T_{gb}^{(1)}$$